INCH-POUND
MS27183L
22 October 2021
SUPERSEDING
MS27183K
16 June 2021

DETAILED SPECIFICATION SHEET

WASHER, FLAT (ROUND, STEEL, CADMIUM OR ZINC-NICKEL PLATED) GENERAL PURPOSE

This specification is approved for use by all components and agencies the Department Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following document listed in ASSIST specified in the solicitation FF-W-92.

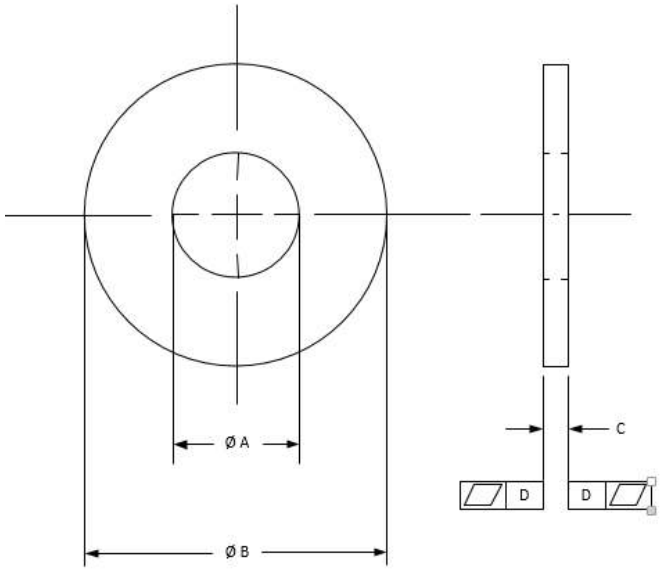


FIGURE 1. Washer, flat.

Comments, suggestions, or questions on this document should be addressed to U.S. Army CCDC, Ground Vehicle Systems Center, ATTN: Standardization, FCDD-GVS-SAT, MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000 or send email to usarmy.detroit.devcom-gvsc.mbx.standardization@mail.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC: 5310

DISTRIBUTION STATEMENT A. Approved for public release, distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T15:17Z
Check the source to verify that this is the current version before use.

TABLE I. Dash numbers and dimensions.

DASH NUMBER	A	TOLERANCE		B	TOLERANCE		C THICKNESS (BASIC)	TOLERANCE		D
		(+)	(-)		(+)	(-)		MAX	MIN	
-1	0.078	0.000	0.005	0.188	0.000	0.005	0.020	0.025	0.016	0.005
-2	0.094	0.000	0.005	0.250	0.000	0.005	0.020	0.025	0.016	0.005
-3	0.125	0.000	0.005	0.250	0.000	0.005	0.022	0.028	0.017	0.005
-4	0.125	0.008	0.005	0.312	0.008	0.005	0.032	0.040	0.025	0.005
-5	0.156	0.008	0.005	0.312	0.008	0.005	0.035	0.048	0.027	0.005
-43	0.156	0.008	0.005	0.250	0.000	0.015	0.016	0.020	0.013	0.005
-6	0.156	0.008	0.005	0.375	0.015	0.005	0.049	0.065	0.036	0.005
-7	0.188	0.008	0.005	0.375	0.015	0.005	0.049	0.065	0.036	0.005
-41	0.188	0.008	0.005	0.438	0.015	0.005	0.049	0.065	0.036	0.005
-8	0.219	0.008	0.005	0.438	0.015	0.005	0.049	0.065	0.036	0.005
-42	0.219	0.008	0.005	0.500	0.015	0.005	0.049	0.065	0.036	0.005
-44	0.125	0.008	0.005	0.375	0.015	0.005	0.032	0.040	0.025	0.005
-45	0.141	0.008	0.005	0.281	0.005	0.005	0.032	0.040	0.025	0.005
-46	0.188	0.008	0.005	0.500	0.015	0.005	0.065	0.080	0.051	0.005
-47	0.219	0.008	0.005	0.365	0.005	0.015	0.032	0.040	0.025	0.005
-48	0.250	0.005	0.020	0.938	0.005	0.015	0.065	0.080	0.051	0.005
-49	0.250	0.015	0.005	0.562	0.015	0.005	0.032	0.040	0.025	0.005
-50	0.250	0.015	0.005	0.625	0.015	0.005	0.032	0.040	0.025	0.005
-9	0.250	0.015	0.005	0.562	0.015	0.005	0.065	0.080	0.051	0.005
-10	0.281	0.015	0.005	0.625	0.015	0.005	0.065	0.080	0.051	0.005
-51	0.250	0.015	0.015	1.000	0.015	0.005	0.065	0.080	0.051	0.005
-52	0.281	0.015	0.005	0.734	0.015	0.007	0.065	0.080	0.051	0.005
-11	0.312	0.015	0.005	0.734	0.015	0.007	0.065	0.080	0.051	0.005
-12	0.344	0.015	0.005	0.688	0.015	0.007	0.065	0.080	0.051	0.005
-13	0.375	0.015	0.005	0.875	0.030	0.007	0.083	0.104	0.064	0.005
-53	0.312	0.015	0.005	0.875	0.015	0.015	0.065	0.075	0.051	0.005
-54	0.312	0.015	0.005	0.500	0.015	0.005	0.035	0.048	0.027	0.005
-55	0.375	0.015	0.005	0.734	0.015	0.005	0.083	0.094	0.064	0.005
-56	0.375	0.015	0.005	0.625	0.030	0.010	0.035	0.048	0.027	0.005
-57	0.406	0.015	0.005	0.875	0.015	0.007	0.063	0.070	0.051	0.005
-58	0.412	0.010	0.005	0.625	0.005	0.005	0.032	0.040	0.025	0.005
-14	0.406	0.015	0.005	0.812	0.015	0.007	0.065	0.080	0.051	0.005
-15	0.438	0.015	0.005	1.000	0.030	0.007	0.083	0.104	0.064	0.005
-59	0.438	0.015	0.005	1.365	0.015	0.007	0.083	0.104	0.064	0.005
-60	0.438	0.015	0.005	0.875	0.015	0.007	0.065	0.080	0.051	0.005
-61	0.438	0.015	0.005	1.625	0.015	0.007	0.065	0.104	0.051	0.005
-62	0.469	0.005	0.020	0.734	0.030	0.007	0.065	0.080	0.051	0.005
-16	0.469	0.015	0.005	0.992	0.015	0.007	0.065	0.080	0.051	0.005
-17	0.500	0.015	0.005	1.250	0.030	0.007	0.083	0.104	0.064	0.005
-18	0.531	0.015	0.005	1.062	0.030	0.007	0.095	0.121	0.074	0.005

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T15:17Z
Check the source to verify that this is the current version before use.

TABLE I. Dash numbers and dimensions - Continued.

-19	0.562	0.015	0.005	1.375	0.030	0.007	0.109	0.132	0.086	0.005
-20	0.594	0.015	0.005	1.156	0.030	0.007	0.095	0.121	0.074	0.005
-63	0.656	0.015	0.010	2.125	0.030	0.007	0.095	0.115	0.080	0.005
-21	0.656	0.030	0.007	1.312	0.030	0.007	0.095	0.121	0.074	0.005
-22	0.688	0.030	0.007	1.750	0.030	0.007	0.134	0.160	0.108	0.005
-23	0.812	0.030	0.007	1.469	0.030	0.007	0.134	0.160	0.108	0.005
-24	0.812	0.030	0.007	2.000	0.030	0.007	0.148	0.177	0.122	0.005
-25	0.938	0.030	0.007	1.750	0.030	0.007	0.134	0.160	0.108	0.010
-26	0.938	0.030	0.007	2.250	0.030	0.007	0.165	0.192	0.136	0.010
-27	1.062	0.030	0.007	2.000	0.030	0.007	0.134	0.160	0.108	0.010
-28	1.062	0.030	0.007	2.500	0.030	0.007	0.165	0.192	0.136	0.010
-29	1.250	0.030	0.007	2.750	0.030	0.007	0.165	0.192	0.136	0.010
-30	1.375	0.030	0.007	3.000	0.030	0.007	0.165	0.192	0.136	0.010
-31	1.500	0.045	0.010	3.250	0.045	0.010	0.180	0.213	0.153	0.010
-32	1.625	0.045	0.010	3.500	0.045	0.010	0.180	0.213	0.153	0.010
-33	1.750	0.045	0.010	3.750	0.045	0.010	0.180	0.213	0.153	0.010
-34	1.875	0.045	0.010	4.000	0.045	0.010	0.180	0.213	0.153	0.010
-35	2.000	0.045	0.010	4.250	0.045	0.010	0.180	0.213	0.153	0.010
-36	2.125	0.045	0.010	4.500	0.045	0.010	0.180	0.213	0.153	0.010
-37	2.375	0.045	0.010	4.750	0.045	0.010	0.220	0.248	0.193	0.010
-38	2.625	0.045	0.010	5.000	0.045	0.010	0.238	0.280	0.210	0.010
-39	2.875	0.065	0.010	5.250	0.065	0.010	0.259	0.310	0.228	0.010
-40	3.125	0.065	0.010	5.500	0.065	0.010	0.284	0.327	0.249	0.010

REQUIREMENTS

1. MATERIAL, STEEL, CARBON GRADES 1005 THRU 1020 (UNS G10050 THRU UNS G10200), IN ACCORDANCE WITH ASTM A108, ASTM-A109/A109M, OR ASTM A1011/A1011M, ANY TEMPER.

2. PROTECTIVE COATING AND SURFACE TREATMENTS: ZINC-NICKEL IN ACCORDANCE WITH SAE AMS2417 CADMIUM PLATE /1 IN ACCORDANCE WITH OR SAE-AMS-QQ-P-416, TYPE II, CLASS 3.

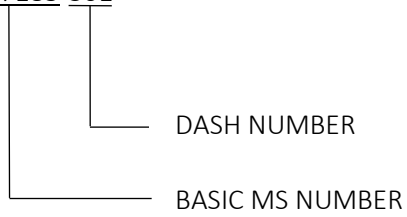
/1 Cadmium is an R1, R2, and D1 substance in the Aerospace and Defense Declarable Substance List. Zinc-Nickel plate per SAE AMS2417 is the preferred method.

3. SURFACE TEXTURE, BEARING SURFACE ROUGHNESS SHALL NOT EXCEED 125 MICROINCHES (Ra) IN ACCORDANCE WITH ANSI/ASME B46.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T15:17Z
Check the source to verify that this is the current version before use.

4. PART NUMBER, THE PART NUMBER SHALL CONSIST OF THE BASIC MS NUMBER FOLLOWED BY A DASH NUMBER FROM TABLE I AND PLATING SUFFIX (IF REQUIRED).

EXAMPLE MS27183-501



EXAMPLE PART NUMBERS:

MS27183-1 - WASHER, FLAT (ROUND STEEL), ID - .078, OD - .188, THICKNESS .020, CADMIUM PLATED.
MS27183-1QX - WASHER, FLAT (ROUND STEEL), ID - .078, OD - .188, THICKNESS .020, ZINC-NICKEL, PLATED

TABLE II. PART NUMBERS.

PART NUMBERS				
CANCELLED				PREFERRED
UNCOATED	PHOSPHATE	CAD OR ZN	CADMIUM	Zinc-Nickel
MS15795-1	MS15795-101	MS15795-201	MS27183-1	MS27183-501
-2	-102	-202	-2	-502
-3	-103	-203	-3	-503
-4	-104	-204	-4	-504
-5	-105	-205	-5	-505
-6	-106	-206	-6	-506
-7	-107	-207	-7	-507
-8	-108	-208	-8	-508
-9	-109	-209	-9	-509
-10	-110	-210	-10	-510
-11	-111	-211	-11	-511
-12	-112	-212	-12	-512
-13	-113	-213	-13	-513
-14	-114	-214	-14	-514
-15	-115	-216	-15	-515
-16	-116	-216	-16	-516
-17	-117	-217	-17	-517
-18	-118	-218	-18	-518
-19	-119	-219	-19	-519
-20	-120	-220	-21	-521
-21	-121	-221	-22	-522
-22	-122	-222	-23	-523
-23	-123	-223	-24	-524
-24	-124	-224	-25	-525
-25	-125	-225	-26	-526
-26	-126	-226	-27	-527
-27	-127	-227	-28	-528
-28	-128	-288	-29	-529
-29	-129	-229	-30	-530
-30	-130	-230	-31	-531
-31	-131	-231	-32	-532
-32	-132	-232	-33	-533
-33	-133	-233	-34	-534
-34	-134	-234	-35	-535
-35	-135	-235	-36	-536
-36	-136	-236	-37	-537
-37	-137	-237	-38	-538
-38	-138	-238	-39	-539
-39	-139	-229	-40	-540
-40	-140		-20	-520

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T15:17Z
Check the source to verify that this is the current version before use.

NOTES:

1. ALL DIMENSIONS ARE IN INCHES.
2. IN THE EVENT OF A CONFLICT BETWEEN THE TEXT OF THIS DOCUMENT AND THE REFERENCES CITED HEREIN, THE TEXT OF THIS DOCUMENT SHALL TAKE PRECEDENCE
3. UNLESS OTHERWISE SPECIFIED, ISSUE OF REFERENCED DOCUMENTS ARE THOSE IN EFFECT AT THE TIME OF SOLICITATION
4. CADMIUM OR ZINC COATED STEEL WASHERS ONLY, PREVIOUSLY SPECIFIED ON MS15795 WERE CANCELLED AND SUPERSEDED BY CADMIUM OR ZINC-NICKEL PLATED STEEL WASHERS SPECIFIED HEREIN
5. USE THE PART NUMBERS GIVEN IN TABLE II
6. IN THE PRECEDING SHEET OF THIS STANDARD, THE CANCELLED WASHERS CANNOT ALWAYS BE REPLACED WITH LIKE FINISHES
7. EXHISTING STOCKS OF WASHERS SHOULD BE USED UNTIL EXISTING STOCK IS DEPLETED
8. USE ZINC-NICKEL WASHERS FOR NEW DESIGN AND REPLACEMENT
9. THE MARGINS OF THIS SPECIFICATION ARE MARKED WITH VERTICAL LINES TO INDICATE WHERE CHANGES FROM THE PREVIOUS ISSUE WERE MADE. THIS WAS DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE PREVIOUS ISSUE.

CUSTODIANS
ARMY – AT
NAVY – OS
DLA – CQ

PREPARING ACTIVITY
ARMY – AT

(PROJECT 5310-2021-026)

NOTE: The activities listed above were interested in this document as of the date of document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T15:17Z
Check the source to verify that this is the current version before use.